

 इजानेन प्रकाशते जगत् INDUS UNIVERSITY				INDUS INSTITUTE OF TECHNOLOGY & ENGINEERING Constituent Institute of Indus University				
Subject: Internet of Things								
Program: B. Tech CE/CS/IT				Subject Code: CE0622			Semester: VI	
Teaching Scheme (Hours per week)				Examination Evaluation Scheme (Marks)				
Lecture	Tutorial	Practical	Credits	University Theory Examination	University Practical Examination	Continuous Internal Evaluation (CIE)- Theory	Continuous Internal Evaluation (CIE)- Practical	Total
3	0	2	4	40	40	60	60	200

Course Objectives

1. Introduce evolution of internet technology and need for IoT.
2. Train the students to build IoT systems using sensors, single board computers and open source IoT platforms that help in skill development.
3. To identify the design, development and security challenges in IoT Systems.
4. To study IoT Applications in Different Domains and be able to measure their performance that enhances the employability skills of students.
5. To implement basic IoT Applications on Embedded Platforms to enhance entrepreneurship skills in students.

CONTENTS

UNIT-I

[12 hours]

Introduction to IoT

Defining IoT, Characteristics of IoT, Physical design of IoT, Logical design of IoT, Functional blocks of IoT, Communication models & APIs, Machine to Machine, IoT versus Machine to Machine, Challenges in IoT: Design challenges, Development challenges, Security challenges

Application of IoT: Home automation, Industry applications, Surveillance applications

UNIT-II

[12 hours]

IoT Reference Architecture- Getting Familiar with IoT Architecture, Various architectural views of IoT such as Functional, Information, Operational and Deployment.

Constraints affecting design in IoT world- Introduction, Technical design Constraints.

Web Infrastructure for managing IoT Resources: Introduction, Open IoT Architecture for IoT/Cloud Convergence, Scheduling Process and IoT Service Lifecycle, Device/Cloud Collaboration Framework

UNIT-III

[12 hours]

Internet of Things Privacy, Security and Governance: Introduction, Overview of Governance, Privacy and Security Issues, Contribution from FP7 Projects, Security, Privacy and Trust in IoT-Data-Platforms for Smart Cities, First Steps Towards a Secure Platform, Smartie Approach. Data Aggregation for the IoT in Smart Cities and Security.

UNIT-IV

[12 hours]

PREPARING IOT PROJECTS: Interoperability in IoT, Introduction to Arduino Programming, Integration of Sensors and Actuators with Arduino, Introduction to Python programming, Introduction to Raspberry Pi, Implementation of IoT with Raspberry Pi.

Course Outcomes

At the end of this subject, students should be able to:

1. Explain the Principles of Internet of Things
2. Design and develop IoT based sensor systems.
3. Employ IoT Solutions to Real Time Engineering Problems
4. Familiar with the Data Management Techniques, Architectures and various key enablers to enable practical IoT systems
5. Identify the Challenges and Research Scope in Communication Protocols used in IoT Applications.
6. Solve IoT security problems using light weight cryptography

Text Books:

1. Internet of Things Principles and Paradigms, Edited By Rajkumar Buyya, Amir Vahid Dastjerdi, Morgan Kaufmann, ELSEVIER
2. Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stamatis Karnouskos, Stefan Avesand, David Boyle, “From Machine to Machine to Internet of Things”, Elsevier Publications, 2014

Reference Books:

1. Fundamentals of Wireless Sensors Networks Theory and Practice, Waltenegus Dargie and Christian Poellabauer, WILEY Series
2. Rethinking the Internet of Things A Scalable approach to connecting everything, Francis daCosta, Apress Open
3. Arduino Cookbook, Michael Margolis, O'Reilly
4. Internet of Things – From Research and Innovation to Market Deployment, Edited By Ovidiu Vermesan and Peter Friess, River Publishers

Web Resources:

1. NPTEL Lecture: <https://nptel.ac.in/courses/106105166/>

LIST OF EXPERIMENTS

Experi- ment. No.	Title	Learning Outcomes
1	Familiarization with Arduino/Raspberry Pi and perform necessary software installation.	CO-1
2	To interface Push button/Digital sensor (IR/LDR) with Arduino/Raspberry Pi and write a program to turn ON LED when push button is pressed or at sensor detection.	CO-2,3
3	Interface analog sensor (PIR Sensor, temperature sensor LM35, Ultrasonic Sensor) with - Arduino	CO-2
4	To interface Bluetooth with Arduino/Raspberry Pi and write a program to turn LED ON/OFF when '1'/'0' is received from a smart phone using Bluetooth.	CO-2,3
5	To install MySQL database on Raspberry Pi and perform basic SQL queries.	CO-3,4
6	Write a program on Arduino/Raspberry Pi to publish temperature data to MQTT broker.	CO-1,3

7	Write a program on Arduino/Raspberry Pi to retrieve temperature and humidity data from thingspeak cloud.	CO-3,4
8	Write a program to create a TCP server on Arduino/Raspberry Pi and respond with humidity data to the TCP client when requested.	CO-4,5
9	Creating a webpage and display the values available through Arduino	CO-4,5
10	Open Ended Experiment-Mini Project: Working on any IOT Application	CO-5,6